

Lab 6 Practice: Dual FPGA Boards

Objective

In this practice, you will gain hands-on experience connecting and communicating between two FPGA boards, one acting as the Master, and the other as the Slave. By completing this lab, you will:

1. Learn how to interface two FPGA boards using general-purpose I/O pins.
2. Understand and manage FPGA I/O constraints files (.xdc).

Descriptions

In this practice, you will connect two FPGA boards using DuPont wires (杜邦線). One board will be designated as the **Master** and the other as the **Slave**. The Master board reads switch inputs and transmits corresponding data to the Slave board, which drives its LEDs accordingly.

Achieve the following objectives:

1. Initially, all the LEDs on the **Slave** FPGA are turned on.
2. Configure the LEDs on the **Slave** FPGA board to reflect the status of the switches on the **Master** FPGA board. Normally, only one switch will be active (pulled up). For example, if the SW0 (V17) on the **Master** board is pulled up, the LED0 (U16) on the **Slave** board should turn OFF.
3. However, if more than one switches are pulled up simultaneously, only the **lowest-indexed switch** determines which LED on the **Slave** is turned OFF (e.g., if both SW2 and SW7 are active, only LED2 **should be darkened**). (11/12 update)
4. Use 5 DuPont wires (杜邦線) to connect two FPGA boards.
 - a. Only 4 signal wires are required for communication, since 4 bits can encode up to 16 values.
 - b. Ensure **ground (GND)** is also connected between the two boards.
5. Choose appropriate PMOD ports for the inter-board connections and create your own constraints file (.xdc file) for both the Master and Slave.

Template

Use the following template as the starting point for your design:

Notice: The data_out and data_in I/O ports are the signals used to transmit data between the two FPGA boards.

```
module lab6_practice_master (  
    input wire clk,  
    input wire rst,  
    input wire [7:0] sw, // switches (11/13 update)  
    output wire [3:0] data_out // data (number) to slave  
);  
  
// add your design here  
endmodule  
  
module lab6_practice_slave (  
    input wire clk,  
    input wire rst,  
    input wire [3:0] data_in, // data (number) from master  
    output wire [7:0] led // LEDs (11/13 update)  
);  
  
// add your design here  
endmodule
```

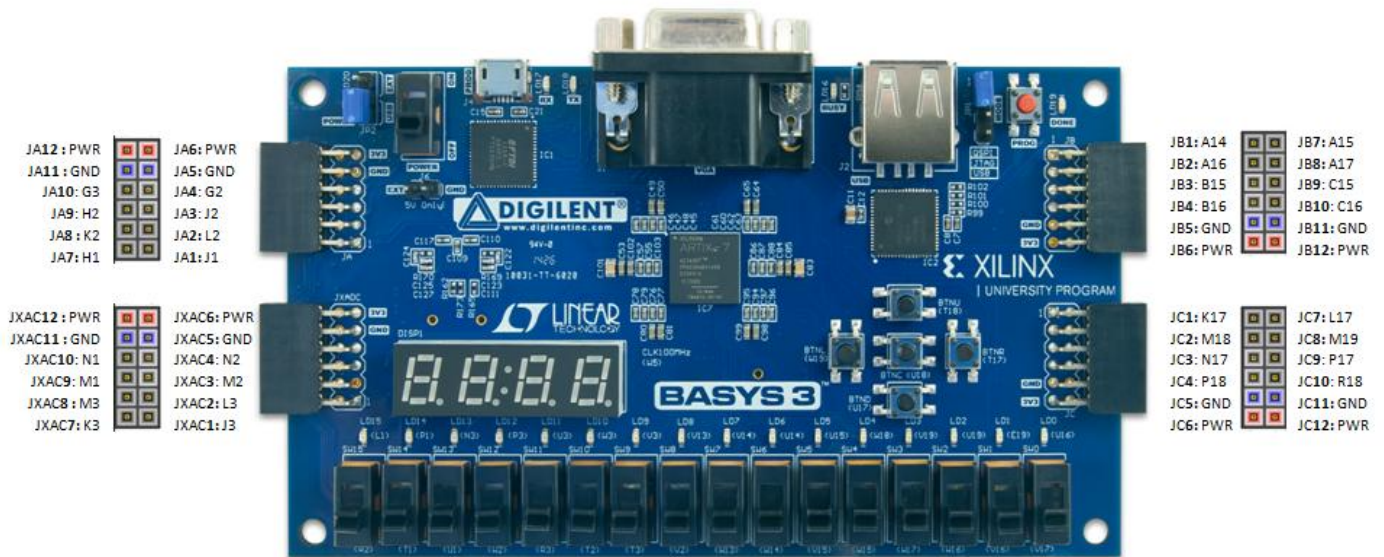
Demo Video

<https://drive.google.com/file/d/1BWN7LxiYKsk0ofQBiKiuIV2a-vpLzkDN/view?usp=sharing>

Hint

- Here is the pin-out diagram for the PMOD connector for your reference:

Basys3: Pmod Pin-Out Diagram



- Here is an example for a revised constraints file:

```

133  ## Pmod Header JA
134  ## Sch name = JA1
135  set_property PACKAGE_PIN J1 [get_ports {data_out[0]}]
136  | set_property IOSTANDARD LVCMOS33 [get_ports {data_out[0]}]
137  # Sch name = JA2
138  set_property PACKAGE_PIN L2 [get_ports {data_out[1]}]
139  | set_property IOSTANDARD LVCMOS33 [get_ports {data_out[1]}]
140  # Sch name = JA3
141  set_property PACKAGE_PIN J2 [get_ports {data_out[2]}]
142  | set_property IOSTANDARD LVCMOS33 [get_ports {data_out[2]}]
143  # Sch name = JA4
144  set_property PACKAGE_PIN G2 [get_ports {data_out[3]}]
145  | set_property IOSTANDARD LVCMOS33 [get_ports {data_out[3]}]

```

Attention

- If you create several modules, merge them into a single Verilog file.
- This practice prepares you for the next lab. Make sure you fully understand your design.
- Feel free to ask questions about the specification on the EECLASS forum.
- DO NOT copy-and-paste code segments from PDFs (hidden characters may cause hard-to-debug syntax errors).